Mathematics II (BSM 121)

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BSM 121

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Power Series

A **power series** in powers of $z - z_0$ is a series of the form

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n = a_0 + a_1 (z - z_0) + a_2 (z - z_0)^2 + \cdots$$

where z is a complex variable, $a_0, a_1, \cdots$ are complex (or real) constants, called the coeflicients of the series, and z_0 is a complex (or real) constant, called the center of the series.

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If $z_0 = 0$, we obtain as a particular case a power series in powers of z :

$$\sum_{n=0}^{\infty} a_n z^n = a_0 + a_1 z + a_2 z^2 + \cdots$$

This generalizes real power series of calculus.

Convergence of a Power Series

Convergence in a Disk: The-geometric series

$$\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + \cdots$$

converges absolutely if $|z| < 1$ and diverges if $|z| \geq 1$ as we have discussed earlier.

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Convergence for Every z: The-power series (which is known as the Maclaurin series of e^z)

$$\sum_{n=0}^{n} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots$$

It absolutely convergent for every z. In fact, by the ratio test. for any fixed z

$$\left| \frac{z^{n+1}/(n+1)!}{z^n/n!} \right| = \frac{|z|}{n+1} \to 0 \quad \text{as } n \to \infty$$

Convergence of Power Series Continue...

Convergence Only at the Center: The following power series converges only at $z = 0$, but diverges for every $z \neq 0$, as we shall show.

$$\sum_{n=0}^{\infty} n! z^n = 1 + z + 2z^2 + 6z^3 + \cdots .$$

In fact, from the ratio test we have

$$\lim_{n\to\infty} \left| \frac{(n+1)! z^{n+1}}{n! z^n} \right| = \lim_{n\to\infty} (n+1)|z| \to \infty \quad \text{as} \quad n \to \infty \quad (z \text{ fixed and } \neq$$

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(b) If a series converges at a point
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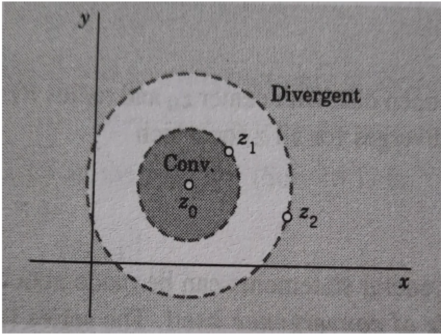

Radius of Convergence of a Power Series

Here, we consider the smallest circle with center z_0 that includes all the points at which a given power series converges. Let R denote its radius. The circle

$$|z - z_0| = R$$

is called the **circle of convergence** and its radius R **the radius of convergence** of the power series.

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Then convergence of power series implies convergence everywhere within that circle, that is, for all z for which

$$|z - z_0| < R$$

(the open disk with center z_0 and radius R). Also, since R is as small as possible, the power series diverges for all z for which

$$|z - z_0| > R$$

Radius of Convergence R

Suppose that the sequence $\left|\frac{a_{n+1}}{a_n}\right|, n = 1, 2, \cdots$, converges with limit L^*, If $L^* = 0$, then $R = \infty$; that is, the power series converges for all z. If $L^* \neq 0$ (hence $L^* > 0$), then

$$R = \frac{1}{L^*} = \lim_{n \to \infty} \left|\frac{a_n}{a_{n+1}}\right|$$

(Cauchy-Hadamard formula)
If $|a_{n+1}/a_n| \to \infty$, then $R = 0$ (convergence only at the center z_0).

Examples: Series with centre at Origin

For what values of x do the following power series converge?

$z_n = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$

Solution:

Apply the Ratio Test to the series $\sum |z_n|$, where z_n is the nth term of the series.

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$$\lim_{n\to\infty} \left| \frac{z_{n+1}}{z_n} \right| = \lim_{n\to\infty} \frac{n}{n+1} |x| \to |x| \quad \text{(show your work here).}$$

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The series converges absolutely for $|x| < 1$.
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At $x = 1$, we get the alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$, which converges.
At $x = -1$ we get $-1 - \frac{1}{2} - \frac{1}{3} - \frac{1}{4} - \cdots$, the negative of the harmonic series; it diverges.
Thus, Series converges for $-1 < x \leq 1$ and diverges elsewhere.

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The value at $x = -1$ is the negative of the value at $x = 1$.
Thus given series converges for $-1 \leq x \leq 1$ and diverges elsewhere.

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These examples illustrates how we usually test a power series for convergence, and the possible results.

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The radius of the convergence of the given power series is

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The series converges in the open disk $|z-3i| < \frac{1}{4}$ of radius $\frac{1}{4}$ and diverge at its boundary. Center of the seires is $3i$.

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$|z-3i| < \frac{1}{4} \implies -1/4 < z-3i < 1/4 \implies 3i - 1/4 < z < 3i + 1/4$

Some Important Series

Taylor series: The Taylor series of $f(x)$ about $x = a$ is,

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Maclaurin series: Taylor series about $a = 0$.

$$f(x) = \sum_{n=0}^{\infty} \frac{f^n(0)}{n!}x^n$$

$$= f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^n(0)}{n!}x^n + R_n$$

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Using the ratio test to determine the convergence set for this power series, we find that the limit of the ratio of consecutive terms, that is

$$L = \lim_{n\to\infty} \left| \frac{\frac{(-1)^{n+2}(x-1)^{n+1}}{n+1}}{\frac{(-1)^{n+1}(x-1)^{n}}{n}} \right|$$

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so the series converges for $L = |x-1| < 1$ or, equivalently, for

$$-1 < x-1 < 1$$

$$\implies 0 < x < 2 \implies \text{Thus, required interval is} \quad (0,2)$$

For the given power series, find the radius of convergence and the interval of absolute convergence.

$$\sum_{n=0}^{\infty} n! x^n$$

Interval and Radius of Convergence

For the given power series, find the radius of convergence and the interval of absolute convergence.

$\sum_{n=0}^{\infty} n! x^n$

Solution: Applying the ratio test, we find that the limit of the absolute value of the ratio of consecutive terms $n!x^n$ and $(n+1)!x^{n+1}$ is

$$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(n+1)!x^{n+1}}{n!x^n}\right| = \lim_{n\to\infty} |(n+1)x|$$
$$= \infty \quad \text{unless } x = 0$$

Therefore, the power series converges only for $x = 0$. The radius of convergence is $R = 0$, and the interval of absolute convergence is the single point $x = 0$.

Taylor's Series Expansion Of $\cos z$ at $z = \frac{\pi}{4}$

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Taylor Series for $f(z)$ is

$$f(z) = f\left(z_0\right) + \frac{f'\left(z_0\right)}{1!}\left(z - z_0\right) + \frac{f''\left(z_0\right)}{2!}\left(z - z_0\right)^2 + \frac{f'''\left(z_0\right)}{3!}\left(z - z_0\right)^3 + \cdots$$

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So, Taylor Series for $\cos z$ about $z = \pi/4$,

$$\cos z = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}(z - \pi/4) - \frac{1}{2!\sqrt{2}}(z - \pi/4)^2 + \frac{1}{3!\sqrt{2}}(z - \pi/4)^3 + \cdots$$

Maclurin Series: Taylor's series at $x = 0$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots (|x| < 1)$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

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Classwork

1. Use direct comparison test to show $\sum_{n=2\infty} \frac{1}{ln(n)}$ diverges. Also use alternating series test to show that series converges.

2. For each of the following power series, find the radius of convergence and the interval of absolute convergence.

a. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ b. $\sum_{n=0}^{\infty} \frac{x^{2n}}{4^n}$ c. $\sum_{n=1}^{\infty} \frac{x^n}{(2n)!}$

d. $\sum_{n=1}^{\infty} \frac{5^n x^n}{n!}$ e. $\sum_{n=0}^{\infty} \frac{n! x^n}{3^n}$

Classwork

3. The series for $\sin x$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!} + \cdots$$

converges to $\sin x$ for all $x \in \mathbb{R}$.

(a) Find the first six nonzero terms of the series for $\cos x$. For what values of x does the series converge?

(b) By replacing x with $2x$ in the series for $\sin x$, find a power series that converges to $\sin(2x)$ for all x.

(c) Using your result from part (a) and series multiplication, calculate the first six nonzero terms of the power series for $2 \sin x \cos x$. Compare your result with your answer from part (b).

Thank You